Recurrent computation and reasoning: a pass-resolved study of Nanbeige with Hermes/TRM controls

Working research draft, 6 September 2026. The reasoning benchmark experiment remains incomplete. Tables below report completed measurements only.

Abstract

Looped language models reuse parameters across successive transformations of a token's hidden state. We develop a reproducible dataset and measurement pipeline for studying this computation in Nanbeige4.2-3B across ARC-AGI-1, ARC-AGI-2, and the logic and math subsets of INTELLECT-3-RL. The corpus contains 34,886 question records, with grading metadata separated from model inputs and cross-release ARC evaluation aliases excluded from calibration. A completed Q4 pilot compares native two-pass execution with a one-pass ablation on 18 inputs. All 36 continuations reach the 2,048-token cap without eligible final answers; both arms score zero on eight evaluation units. Native trajectories yield 900 activation arrays across both passes, with a marked change in pass similarity at physical layer 22. The initial frozen VPD bases cover only about 0.26% of selected weight energy, and broader SAE fits have substantial train/evaluation reconstruction gaps. These measurements localize differences but do not establish an answer-level mechanism.

Separately, we reproduce the Hermes Campsite solver experiment on all 109 source rows. Filtering enumerated candidates with the official verifier, without giving the solver target solutions, yields 109/109 valid answers and 99/109 canonical matches. Historical TRM-augmented predictions achieve 38/109 official-valid answers and 33/109 canonical matches. The historical 27B runs used short output caps and disabled thinking, and their support banks overlap evaluation examples. They therefore do not provide a matched long-reasoning scale comparison. We also identify discrepancies between puzzle wording, reference answers, and official verification. A completed 64-component BF16 fit on saved Q4 trajectories raises selected-matrix weight coverage to about 2.2%, but held-out SAE reconstruction remains inadequate at several passes. The longer-budget comparison, causal answer interventions, and a fresh audited 27B/TRM comparison remain incomplete; no general recurrence advantage, chain-of-thought compression mechanism, or scale law is established.

1. Research question

The primary question is whether additional recurrent transformations improve reasoning outcomes, and which parameter components and sparse features mediate any improvement. Hidden-state movement alone is insufficient: a useful mechanism must connect a localized intervention to a scored behavioral endpoint. A second question concerns how recurrence inside a language model compares with external control implemented by Hermes skills, candidate repair, and TRM-based routing.

Nanbeige is an inspectable model for these experiments. Its publisher describes parameter reuse through a looped transformer architecture. Our pinned configuration and instrumented execution use 22 physical transformer layers twice, with normalization at each loop boundary. This is fixed recurrence. Neither this implementation nor its measurements establish facts about Astra's undisclosed architecture. Nanbeige model and source.

The term “tuck” is treated as a hypothesis requiring an operational definition. One prospective test is whether a second-pass feature change carries task-relevant information that causally improves a later answer while reducing the need for additional generated reasoning. A shorter trace, a smaller activation norm, or a second-pass router count alone would not satisfy this definition. Recursive architectures and external recurrent controllers must be evaluated with their own training and inference budgets. Recursive reasoning with tiny networks.

The Nanbeige report describes diminishing additional gains beyond two passes, weaker results with cross-pass KV sharing, and better performance when recurrence is trained from scratch. It also uses difficulty-aware length penalties during reasoning RL, alternating constrained and free-expansion phases. Thus post-training creates a separate potential cause of shorter reasoning. Our checkpoint ablation cannot separate this training effect from the architecture's contribution; intermediate training checkpoints would be needed. These are publisher-reported findings, not replications in our experiment. Nanbeige technical report, Sections 2.1 and 3.2.3.

2. Sources and dataset

All inputs are pinned by revision and file hash in `sources.lock.json`. The four source lanes are retained separately. INTELLECT examples come from the public RL **training** corpus; their local splits are not an official held-out test and do not establish freedom from model pretraining exposure. ARC evaluation here is public, not private. ARC-AGI-1, ARC-AGI-2, INTELLECT-3-RL.

Lane	Calibration/train	Local dev	Evaluation	Historical evaluation	Quarantine
ARC-AGI-1	415	0	419	0	1
ARC-AGI-2	681	0	167	0	395
INTELLECT logic	8,952	1,145	1,050	109	391
INTELLECT math	16,931	2,068	2,162	0	0

Counts are question records: ARC contributes one record per test input, so these numbers are not counts of independent ARC tasks. Every test input of a task receives the same partition. ARC groups join both task identifiers and hashes of public examples/test inputs across releases; any alias of an evaluation task is excluded from calibration. INTELLECT's 391 embedded ARC examples are quarantined pending a complete mapping. The 109 Campsite examples are reserved for comparison with historical Hermes receipts. Exact question duplication is also checked across partitions; semantic near-duplicate decontamination remains future work.

The model-facing JSONL contains questions, provenance, and grouping identifiers. A separate grader sidecar contains ARC test outputs, INTELLECT math answers, or logic metadata. Neither the sidecar nor historical predictions are included in the GPU inference payload. Demonstration outputs supplied as part of an ARC task are legitimate inputs.

The feasibility subset selects two groups per lane and calibration/evaluation partition by a fixed hash order, with one historical Campsite evaluation case. All test inputs of the selected ARC groups are retained, producing 18 inputs. Selection uses prompt length and family, never gold outcomes; 13 candidate groups exceeded the 2,048-token prompt cap. Consequently this subset cannot estimate full-benchmark performance without selection bias. The full corpus remains available for subsequent runs.

The subsequent benchmark cohort has also been materialized in `benchmark_v1`. It includes every public ARC-AGI-1 evaluation task (400 tasks, 419 test inputs), every public ARC-AGI-2 evaluation task (120 tasks, 167 inputs), 200 fixed logic holdout examples across 27 eligible families, 200 fixed math holdout examples, all 109 historical Campsite cases, and 128 calibration records. Its 1,223 records contain 2,528,898 prompt tokens, with median 929 and maximum 17,660; none is excluded by prompt length. Logic sampling is family-stratified and must be reported as such, while the Campsite historical slice remains separate. Materializing this cohort does not mean it has been evaluated.

3. Measurement methods

3.1 Generation and pass ablation

The planned BF16 experiment compares native two-pass execution with a one-pass ablation of the same checkpoint. Both use greedy decoding, thinking-enabled prompts, and a 2,048-new-token cap. This is a test of the trained model's dependence on its second pass; it is **not** a comparison between independently trained one-pass and two-pass architectures. Equal token caps are not equal compute. Elapsed time, prompt/generated token counts, and executed physical-layer token counts are recorded; the latter are a proxy rather than a FLOP estimate. Unfinished reasoning and token-cap censoring are retained in the denominator and reported explicitly.

Cached decoding is checked against full-prefix execution for both loop counts before generation. For each native continuation, the exact prompt and generated token sequence are replayed without caching. Hooks retain both invocations of all 22 physical layers and loop-boundary normalization. Up to eight positions are retained in each of four regions: prompt suffix and the early, middle, and late portions of generation. Token IDs and absolute positions accompany every activation archive. These regions are positional, not presumed semantic reasoning stages.

The larger-cohort runner supports left-padded batches and durable resume records, with a separately recorded default cap of 8,192 new tokens. It avoids allocating vocabulary logits for unused prefill positions during generation; that optimization is disabled for losses and parameter decomposition. Batched decoding, final-position logit parity, interruption recovery, and complete-task ARC scoring are checked. Every resume contract includes question, checkpoint, and decoding hashes. Completed batches are skipped; an interrupted batch is regenerated at its original composition, and already-saved trajectories must match before missing records are appended. Replayed work is included in batch-attempt receipts. A timed pause is recorded as incomplete, never as a successful benchmark completion.

The local behavioral fallback uses the separately pinned Q4_K_M conversion of Nanbeige, with q8_0 key/value caches and a 4,096-token context. It renders the same official HF input token IDs and compares the metadata loop-count settings 2 and 1 using the same quantized physical weights. Greedy decoding, a 2,048-token continuation cap, repeat penalty 1, and disabled context shifting are fixed across conditions. This is a quantized-model experiment; BF16 behavioral parity has not been established. Runtime source confirms physical weight sharing and separate logical KV indices across passes. The full command, DLL/executable hashes, settings returned by the server, stop reasons, and exact output tokens are retained. Laptop thermal throttling affects elapsed time, which is not a controlled compute comparison. Pinned runtime implementation.

Direct C API callbacks also support activation capture from this same Q4 model. A completed pretrained CPU preflight observes 23 arrays under one pass and 46 under two passes on 32 tokens. Capturing leaves the full next-token logit vector exactly unchanged in both cases. All 23 first-pass sites are also exactly invariant

to enabling the second pass in this control. These checks establish callback and pass-index behavior, not reasoning accuracy. Completed generated-trajectory replay records conditional greedy agreement because chunked replay and single-token generation can differ numerically; Section 4.7 reports the observed mismatches.

3.2 Parameter components and shared sparse dictionaries

We adapt Goodfire's adVersarial Parameter Decomposition (VPD), using the core mathematics from its compact `nano_param_decomp` implementation. VPD decomposes parameters; a sparse autoencoder (SAE) instead learns a dictionary of activation features. Our adapter keeps one physical set of parameter factors shared across recurrent invocations while assigning separate invocation caches and intervention masks. The retained upstream source sections, source SHA256 and MIT license are recorded in `experiments/nanbeige_looped/vpd_backend_provenance.json` and `VPD_BACKEND_LICENSE` in the workspace. This is an adaptation to selected matrices in a recurrent model, not a replication of the original whole-model decomposition. VPD method, upstream implementation.

For each selected linear map, the fitted product `V @ U` has the orientation of the transposed PyTorch weight. The residual is `W - (V @ U).T`; retaining it preserves the original map algebraically when all components and the residual are included. Reported component coverage concerns the fitted product alone. Joint calibration optimizes four terms: parameter faithfulness, importance minimality, stochastic reconstruction and adversarial reconstruction. The importance predictor is auxiliary calibration machinery; it does not make Nanbeige's inference loop adaptive. Our short optimization budgets and low fitted coverage limit what can be inferred from an individual component.

At the MLP down projections of physical layers 12 and 22, the decomposition assigns one V/U component basis to each shared weight matrix. Invocation-specific calibration state and intervention masks distinguish the passes. The initial calibration used eight components, 50 warm-up updates, and 50 updates with the four VPD objectives. The completed generated-prefix run uses 64 components with the same update counts. A 64-component fit should not be presumed adequate merely because it is larger; reconstruction and intervention stability must be measured.

For a component contribution D_c, we compare additions αD_c at the first invocation, the second invocation, and both. Applying the same contribution twice preserves sharing in real arithmetic; materializing a BF16 matrix edit can round differently. Single-pass interventions intentionally break sharing to localize propagation. We retain positive and negative doses, central differences, and the vector interaction

$$I_c(\alpha)=h_{both}(\alpha)-h_{first}(\alpha)-h_{second}(\alpha)+h_0.$$

The first-pass effect has more downstream transformations before the final endpoint. Different input states, downstream depth, finite-dose nonlinearity, and BF16 rounding must therefore be separated before interpreting an asymmetry as a specialized reasoning role.

At each selected capture site, one top-k SAE is fitted on balanced training rows from both passes and frozen across all conditions. Feature changes are actual nonlinear re-encodings, $\Delta z=E(h+\Delta h)-E(h)$, not decoder projections of activation differences. Reconstruction is evaluated separately by pass. Feature IDs have no validated semantic interpretation at this stage. The prepared BF16 run retains full generated contexts for capture and effect evaluation. Its VPD fit now selects unpadded prefixes ending at 25%, 50%, and 100% of each training continuation, retaining every preceding prompt token. It shares physical factors and the invocation-aware calibration network across lengths, with separate persistent adversarial state for each length. A small random-weight fixture using the publisher's recurrent architecture passed actual

alternating-length optimization and mask-restoration checks. The first generated-prefix attempt exhausted GPU memory before completion and saved no fitted decomposition; the corrected completed run is reported in Section 4.9. Earlier completed arithmetic results used their original fixed prefix and are unchanged.

The Q4 replay additionally retains inputs to the physical layer-12 and layer-22 MLP down projections at both invocations. A completed 16-token CPU control captured these four extra input arrays with unchanged logits. The completed transfer analysis computes coordinates $c_p = x_p V$ and basis outputs $c_p U$ using the earlier frozen BF16 VPD factors. This is a transfer diagnostic across weight precision, not a newly fitted Q4 decomposition, a causal intervention, or coverage of the omitted residual. The low coverage of the original eight-component bases remains a limitation. New Q4 SAEs were fitted on training trajectories from the completed initial run, with held-out reconstruction reported separately.

A separate cross-precision calibration has now completed and passed independent verification: BF16 Nanbeige receives the exact native two-pass Q4 trajectories from the completed initial pilot, comprising eight training and ten held-out inputs with 49,987 total tokens. All 18 continuations are budget-censored. This VPD uses 64 components at physical layers 12 and 22, 50 faithfulness warm-up updates and 50 joint updates, followed by shared-SAE fitting and signed pass-specific interventions. Full conditioning is retained at the three training-prefix fractions above. Source tokens, split assignments and generation hashes are checked before bundling; remote checkpoint assets must match all 12 pinned file hashes. This is teacher-forced BF16 analysis of Q4-generated trajectories, not BF16 self-generation, Q4 parameter fitting, or an answer-level accuracy experiment. It uses no KV cache and does not relax the separate cached-generation equivalence checks. The original specification is in `cross_precision_vpd_inputs_v1/contract.json`. That attempt logged its first update, then failed while allocating another 1.51 GiB for the adversarial KL computation on a 39.49 GiB GPU, with 38.51 GiB in use by the process. Exact returned input trajectories and cleanup were verified; no completed fitted factors were saved. The revised `cross_precision_vpd_inputs_v2/contract.json` preserves the data, objectives, prefix lengths and update counts while offloading tensors saved for backward to CPU RAM. It introduces no forward recomputation. Small CPU and actual CUDA controls reproduced all losses and 28 parameter/gradient arrays exactly over three varying-length updates. The pretrained retry completed all 50 joint updates, with peak allocated CUDA memory of 23,893,116,928 bytes. The first full-pretrained adversarial loss nevertheless differs between the separate original and retry pods (0.26720446 versus 0.26673719, about 0.175%); the other three first-step objective values agree. We have not isolated the cause and do not claim full-model bitwise equivalence. Completed decomposition and held-out measurements are reported in Section 4.9.

3.3 Behavioral grading

ARC predictions require a rectangular JSON grid of integers 0-9 and exact equality with the target. Both per-input accuracy and all-test-input task success are reported. This feasibility run has one greedy attempt and does not estimate pass@2. Logic uses pinned task-specific verifiers, including explicitly retained task families rather than the environment's default skip list. A documented adapter replaces their Python `eval` lookups with bounded literal/arithmetic parsing: function calls, attribute access, and comprehensions are rejected. This preserves the tested expected answer forms without executing generated Python; it is an adapter, not byte-for-byte official execution. Math uses the official boxed-answer extraction and `math-verify` equivalence with pinned dependencies. On Windows, an outer process timeout replaces the library's incompatible internal multiprocessing timeout; answer extraction and equivalence logic remain unchanged. Infrastructure grading errors are recorded separately from incorrect answers. Official environment source.

4. Completed results

4.1 Pretrained implementation calibration

The completed initial experiment used 12 training and four evaluation arithmetic prompts, with thinking disabled and the final 16 prefill positions retained. The retained token IDs were identical across examples, although preceding questions differed. On this narrow suffix, SAE explained variance ranged from 91.87% to 99.12% across the measured sites and passes. These figures are not evidence of broad semantic feature quality.

Physical layer	Component	First-pass effect norm	Second-pass effect norm	Both-pass effect norm	Interaction norm
12	0	32.882	17.140	32.946	23.964
12	1	33.830	17.604	33.928	24.623
22	0	23.764	0.615	23.764	0.769
22	1	25.324	1.005	25.331	1.257

These are pooled final-normalization activation norms at dose 0.05, not task scores. For the two layer-22 components, the first-pass effect is approximately 25-39 times the second-pass effect. The selected eight-component bases reconstruct only 0.268% and 0.260% of squared weight energy. The remainder stays in the residual component. The calibration passed exact adapter-logit parity and second-pass intervention causality checks; 74 original artifact files were hash-verified. See `../nanbeige_looped/PRETRAINED_RESULTS.md` and its linked manifests.

A subsequent independent check reloaded the dictionaries and verified all 16 saved intervention records, including 1,632 pass-difference, finite-difference, interaction and causal-order calculations and 672 nonlinear SAE re-encodings. It also reproduced all six held-out SAE quality measurements. This strengthens saved-artifact verification without rerunning the model or changing the narrow prefill scope. The receipt is `./nanbeige_looped/prime-20260906T034844Z-719/retrieved/artifacts/nanbeige_looped/bf16_multipass_calibration/independent_verification.json`.

4.2 Hermes solver and historical TRM audit

We pinned the supplied Hermes-Skills repository and recovered all 109 corresponding historical Campsite rows by matching target grids. Targets were used for this retrospective matching and final exact scoring, never supplied to candidate generation. We reran its public enumerator with only the original grid and row/column constraints. It reproduced 92 unique, 16 two-solution, and one four-solution case under its relaxed rule contract. Pinned Hermes source.

Arm on the same 109 cases	Canonical target match	Official verifier accepted
Historical vanilla 27B predictions	0/109	0/109
Historical generic skill	1/109	1/109
Historical logic skill	0/109	0/109
Historical logic skill + TRM	33/109	38/109
Replayed public solver, first candidate	96/109	106/109
Public solver, first candidate passing official verifier	99/109	109/109

The last arm is a new deterministic control: it filters the enumerated candidates using an official verifier object containing only public puzzle inputs. It does not use the canonical target to select a candidate. It establishes complete verifier acceptance for this 109-row slice. It does not establish that a neural TRM alone solved all INTELLECT logic, all math, or ARC. Historical model receipts are not a fresh, compute-matched 27B comparison. Recovered configurations identify Qwen3.5-27B Q4_K_M with 96 output tokens for logic and 32 for numeric-only math, whereas the new Nanbeige run allows 2,048. Thinking settings, retrieval exposure, and controller costs must also be reconciled before attributing a scale advantage.

For the separate 200-row historical math slice, independent string-exact recomputation agrees with the receipt: vanilla 17/200, generic skill 14/200, math skill 12/200, and routed math-skill/TRM final 14/200. The first three arms record 200 model calls each; the TRM arm records 400. The available receipt therefore supports a low-scoring, tightly constrained 27B baseline, but does not itself demonstrate a complete math-benchmark clear. It should not be compared directly with boxed-answer symbolic equivalence scores without regrading and aligning task identities.

A subsequent task-identity audit finds that all 200 ordered historical labels match the first 200 rows of the pinned public math source. The 25 historical observations recoverable from support banks also match those source prompts exactly, with no contradictions and no overlap with the larger cohort's planned calibration records. The other 175 historical prompt identities remain provisional because the original trajectory file is missing. This evidence strengthens the candidate mapping but does not make the output budgets or task exposure comparable.

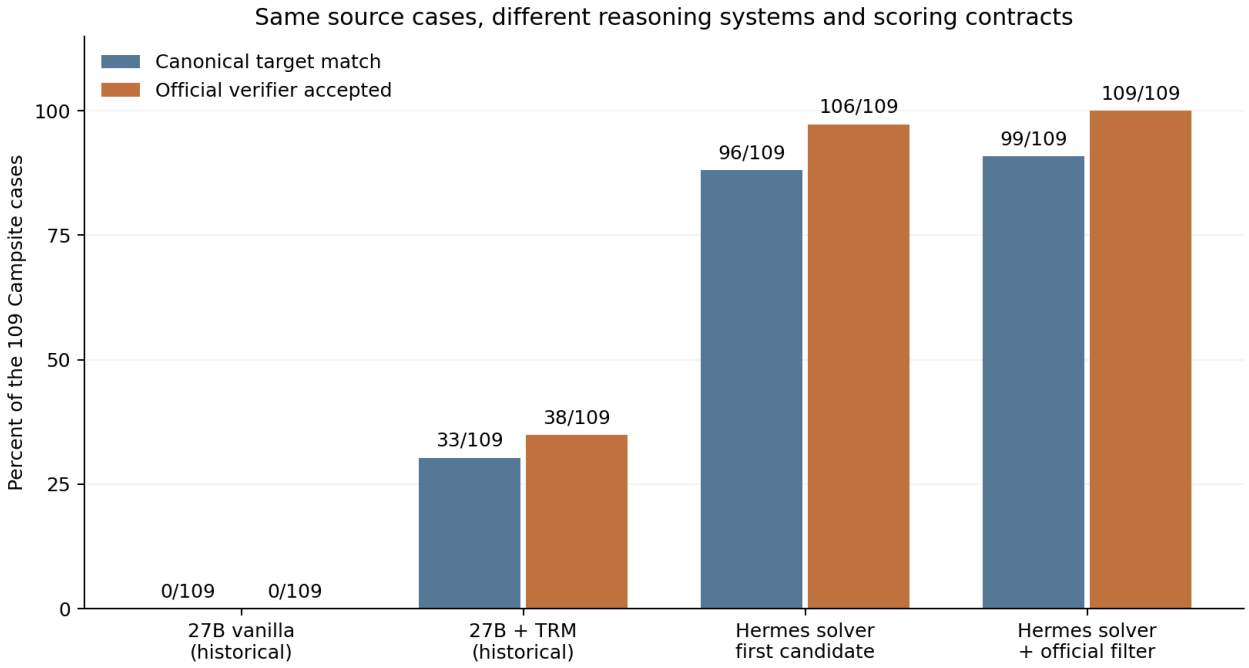
An archived support-bank audit matches public Campsite grids and row/column constraints for 100 historical questions. Of these, 77 pass the retriever's bucket and supervision-weight filters; all 100 enter the archived critic's source bank. Historical logic/TRM outputs pass the official verifier on 21/77 retriever-member cases and 17/32 remaining cases, versus zero vanilla successes in either partition. The latter 32 cases are not a clean support holdout: 23 still appear in the critic's source bank. These counts describe overlap, not which examples reached a particular model call.

The six eligible math support entries match six evaluated prompts and their reference answers. All six historical TRM receipts contain a retrieved answer equal to the reference, although exemplar IDs are absent. On these six rows, vanilla, generic skill, math skill, and routed TRM score 3, 1, 4, and 6 exact matches respectively. On the remaining 194 rows, they score 14, 13, 8, and 8. Thus the aggregate two-answer advantage of routed TRM over the math skill occurs entirely within this support-member partition. Support selection was not randomized, so these partitions do not estimate a causal retrieval benefit. Archived runner code contains normalized-observation exclusion filters, but execution of that exact code for these receipts

has not been established. The bank files and audit source are hashed; the audit does not claim that all historical runtime provenance is recovered.

Inspection of the archived hybrid runners further constrains the meaning of the TRM label in these receipts. The logic loader constructs a retrieval bank, a nearest-neighbor critic, and a constraint/exact-search corrector; the math loader constructs a retrieval bank and nearest-neighbor critic. These auxiliary paths do not load a neural recurrent checkpoint. This statement concerns the inspected historical paths, not every system called TRM in the broader Hermes research. Source hashes and entry points are recorded in `hermes_audit/controller_implementation_audit.json`.

The larger benchmark remains fixed. `hermes_audit/benchmark_support_exclusions.json` identifies its 100 public inputs that overlap archived critic/source support, for exclusion from support banks in a fresh comparison. It does not alter either the benchmark or historical receipts. `retrieval_exposure_summary.json` and its row records preserve the overlap partitions, and their totals reproduce every previously audited historical arm.



Canonical agreement and official acceptance on the 109 Campsite cases

4.3 A scoring-contract discrepancy

The Campsite prompt literally requires each tent to touch exactly one tree. Only 9/109 supplied targets satisfy that condition, while all 109 pass the official verifier's one-to-one matching test. The Hermes enumerator instead admits any tent adjacent to at least one tree; its first candidates pass that relaxed check on 109/109 cases. Enumerating and filtering all candidates finds a literal-contract solution for only nine cases, so the discrepancy is not simply a poor first-candidate choice. Under the official verifier, candidate filtering resolves all 109 cases without canonical-answer access.

These are three different contracts. Reporting only exact target agreement would penalize some valid alternatives; reporting only the relaxed checker would admit three first candidates rejected by the official verifier. The primary comparison in this study uses official verifier acceptance, with canonical agreement and prompt-literal validity retained as diagnostic columns.

4.4 Logic reference and formatting audit

Positive-path probes use the first three source records in each of 29 logic families. They serialize supplied answers or solution metadata into task-specific output formats solely to check the graders; these reconstructions are never passed to the model. All three probes pass in 28 families. All 87 empty-answer probes fail. Futoshiki is the exception: after using its specified space-separated row format, the reference grids still fail the inequality checks.

A complete audit of all 714 Futoshiki source records finds that 697 supplied grids violate at least one recorded inequality; only 17 supplied grids pass the verifier. Nine of the ten Futoshiki records selected anywhere in the larger cohort have this issue. This demonstrates inconsistent reference solutions, not that the underlying puzzles are unsatisfiable or that every generated solution will fail. The cohort and scorer are retained, and this family must be reported separately with reference-quality diagnostics. Records and violated coordinates are preserved in `grader_audit/futoshiki_rows.json`. No reference answers have been repaired.

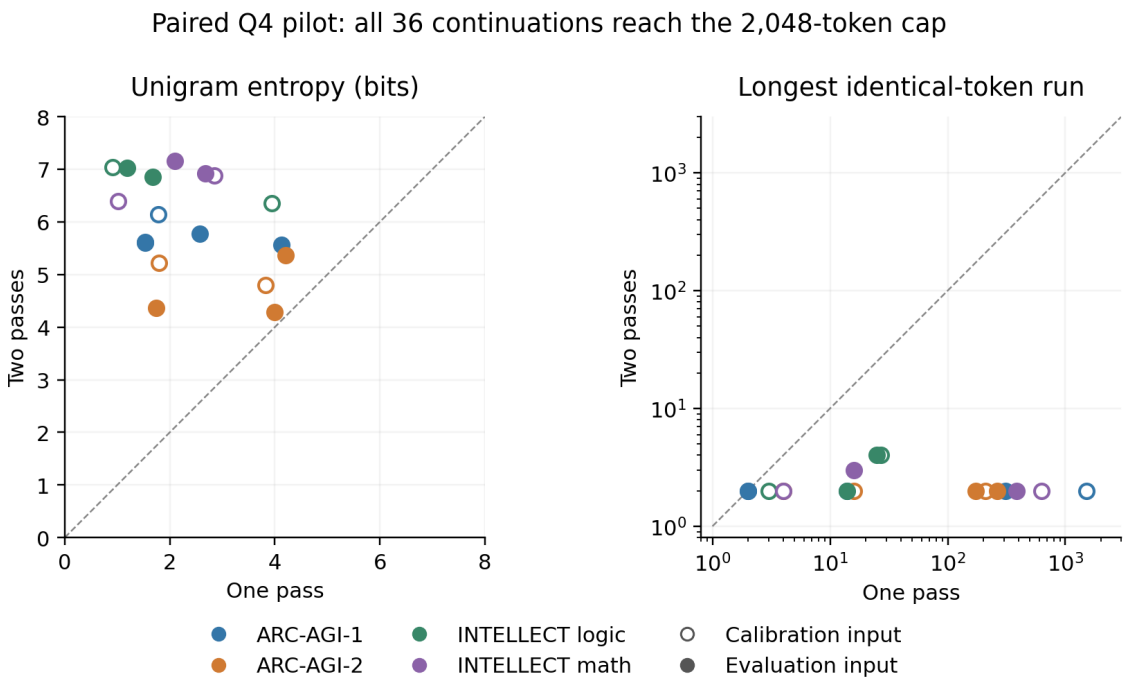
4.5 Completed 2,048-token Q4 pilot outcomes

The paired Q4 pilot completed 36 continuations on 18 public inputs: 73,728 generated tokens in total. All 36 continuations reached the 2,048-token cap, none closed the thinking delimiter, and none supplied an eligible completed final answer. Both conditions therefore scored zero on every evaluation unit, with no grader errors or missing pairs:

Evaluation slice	Units per condition	One pass correct	Two passes correct
ARC-AGI-1	2 complete tasks / 3 test inputs	0	0
ARC-AGI-2	2 complete tasks / 3 test inputs	0	0
INTELLECT logic, local evaluation	1 question	0	0
INTELLECT logic, historical Campsite	1 question	0	0
INTELLECT math	2 questions	0	0

This is an output-censored feasibility result, not evidence of equal underlying reasoning ability or an accuracy benefit from recurrence. ARC is paired at the complete-task level, requiring every test input to succeed; historical evaluation stays separate from the local holdout. Source and score hashes are preserved in `q4_reasoning_pilot_v1/pilot_outcome_summary.json`.

The output-token distributions nevertheless differ. Across all 18 paired inputs, median unigram entropy is 1.95 bits for one pass and 5.96 bits for two passes; the longest identical-token run has medians 20.5 and 2 tokens. Two-pass entropy is higher on all 18 inputs. Among the 10 evaluation input records, entropy medians are 2.34 and 5.70 bits. These are descriptive input-level statistics, not independent task-level significance tests. Structured answers can naturally have low token diversity. Ablating a model trained for two passes can damage general generation; these differences do not identify reasoning compression or a reasoning-specific mechanism.



Paired token entropy and single-token repetition on the completed Q4 pilot

Each point is one input; filled points are evaluation inputs and open points are calibration inputs. Dashed lines indicate equality between pass conditions. Both panels use the same actual 2,048-token continuations. The right panel uses logarithmic axes. No condition produced a completed eligible final answer at this cap.

An exploratory 8,192-token follow-up used the same inputs in a separate artifact directory, with a 10,240-token context and unchanged weight/KV precisions. It saved 17 native two-pass outputs before the next completion request exceeded its 4,096-second HTTP response timeout. The runtime was terminated by the runner's cleanup path. No one-pass output was saved in this follow-up, so the planned 36-output paired comparison, grading and subsequent activation analysis remain incomplete. The 17 durable outputs are retained; an in-flight token count in a server log is not a recovered continuation. No accuracy claim is made from this partial cohort. Initial-budget activation capture, SAE fitting and independent verification remain complete. Local wall time is not a controlled compute comparison: thermal throttling, host-cache restart and experimental pauses affected execution, and the cause of the final throughput slowdown was not isolated. The failure receipt is `runtime_observations/extended_generation_timeout.json`.

One one-pass ARC request exposed a response-transport failure: the server completed 2,048 tokens, but its content parser rejected malformed UTF-8 and returned HTTP 500. That request contributed no recorded data. An actual-runtime control reproduced the failure with a forced byte token and verified that an empty response parser preserves returned token IDs. Both parser settings reproduced a saved 16-token greedy prefix. The failed input was rerun successfully, preserving all earlier raw records and unchanged sampling. Forced token bias was confined to the control; raw token IDs remain authoritative, while decoded text may contain replacement characters. See `parser_failure_receipt.json` and `raw_token_transport_test/verification.json`. The pinned UTF-8 validation is visible in the llama.cpp parser.

BF16 generated-reasoning results remain pending. An A100 attempt stopped before generation when one of 166,144 logits violated the original cache-equivalence tolerance (absolute difference 0.2734375; relative

tolerance 0.02 and absolute tolerance 0.25). A subsequent diagnostic on an RTX 6000 Ada completed and was independently verified against both local checkpoint shards, the frozen source bundle, exact input tokens, and all saved logits. Both BF16 loop counts pass the unchanged full-versus-cached test, with maximum absolute differences 0.046875 and 0.25. FP32 arithmetic on the upcast model reduces those maxima to 0.00000882 and 0.0000610; both pass the predeclared 0.0001 absolute/relative checks, with correct 22/44 cache cardinalities.

Full-versus-cached agreement does not imply cross-precision greedy agreement. In the one-pass condition, BF16 has a zero top-two margin and chooses token 1628, whereas FP32 chooses token 2753 with margin 0.00461. In the native two-pass condition, both precisions choose token 2753. This single 32-token prefix diagnoses numerical behavior; it does not establish benchmark parity. No generation threshold was relaxed. Results and termination are verified in `prime-20260906T191119Z-509/cache_verification.json`. The successful diagnostic pod was terminated; a subsequent BF16-generation launch found no available matching GPU and produced no inference.

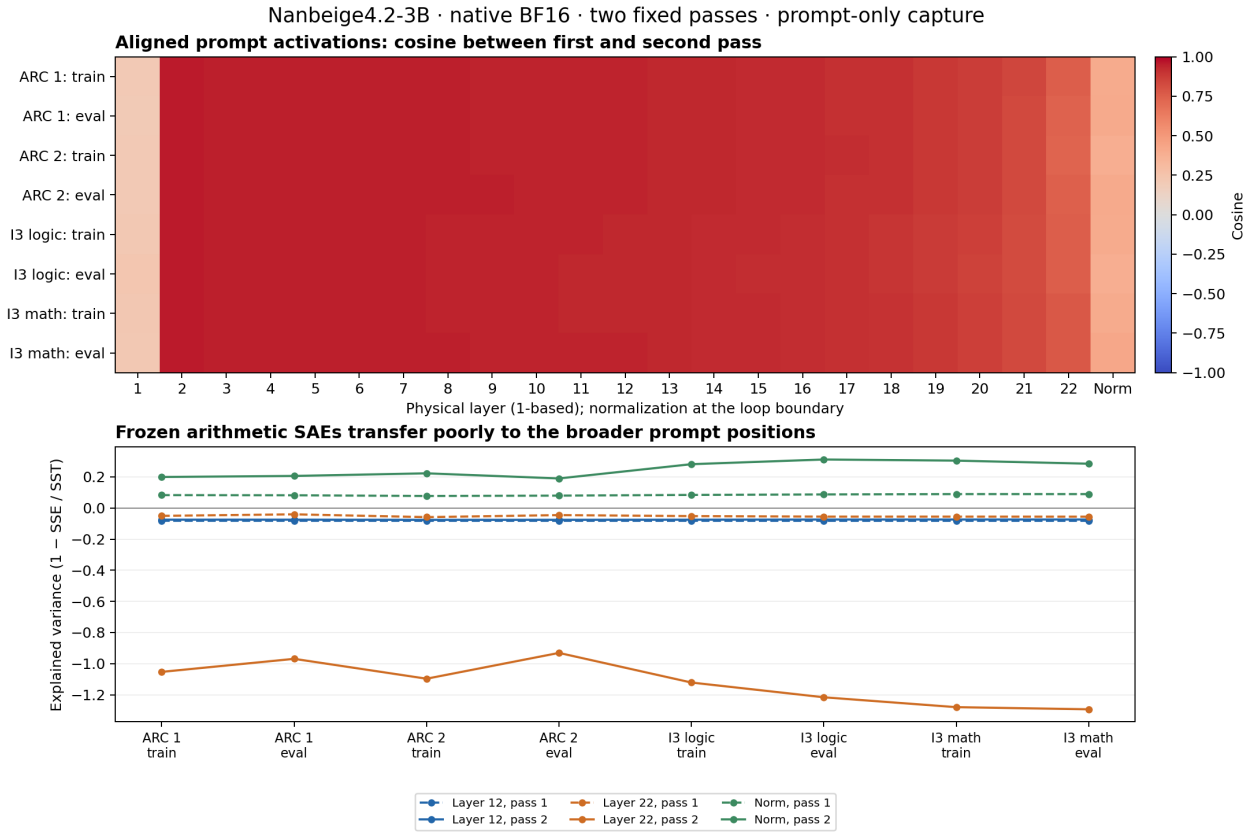
4.6 Completed local prompt capture and frozen SAE transfer

The local fallback completed all eight selected prompts: one calibration and one evaluation example per source lane. It executed the original BF16 checkpoint over 4,208 prompt tokens in 1,757.5 seconds of recorded forward/capture time. Each archive retains the first eight and last 24 positions, yielding 256 retained token positions and 368 finite activation arrays across 23 sites and two invocations. All archive hashes, invocation counts, and array shapes were independently verified. These are the shortest available inputs within each pilot lane/split, not a representative accuracy sample.

The three previously fitted arithmetic SAEs were frozen and applied without new fitting. This produced 24 archives containing both encodings and their actual difference, alongside hidden-state differences for all layers. Reloading each dictionary reproduced its original held-out explained variance within 1.1e-8, ruling out a dictionary-loading error in this control. Across the eight new examples, the reconstruction results were:

Frozen dictionary site	Pass 1 explained variance range	Pass 2 explained variance range
Physical layer 12	-0.0821 to -0.0819	-0.0758 to -0.0737
Physical layer 22	-0.0584 to -0.0404	-1.2935 to -0.9310
Loop-boundary normalization	0.0775 to 0.0897	0.1902 to 0.3120

Explained variance here is 1 minus reconstruction squared error divided by the within-example, within-pass sum of squared deviations from the mean activation. Negative values mean the reconstruction is worse than that mean baseline. These pooled-position results demonstrate poor transfer from the earlier fixed arithmetic suffix to the new combination of beginning and ending prompt positions. Task, token-position, thinking-template, and CPU/GPU execution differences are not separated by this test. The saved feature differences therefore cannot be treated as validated semantic reasoning features. Broader training-only calibration and generated reasoning remain necessary.



Pass similarities and frozen SAE transfer quality across eight prompts

4.7 Generated-trajectory Q4 captures

Native two-pass trajectories from all 18 pilot inputs have now been replayed through the same pinned Q4 checkpoint: 49,987 full-context tokens in 243.7 seconds of measured replay time. The archives retain 576 token positions across prompt, early, middle and late continuation regions. They contain 900 activation arrays: both invocations of all 22 transformer layers and loop-boundary normalization, plus the down-projection inputs at physical layers 12 and 22.

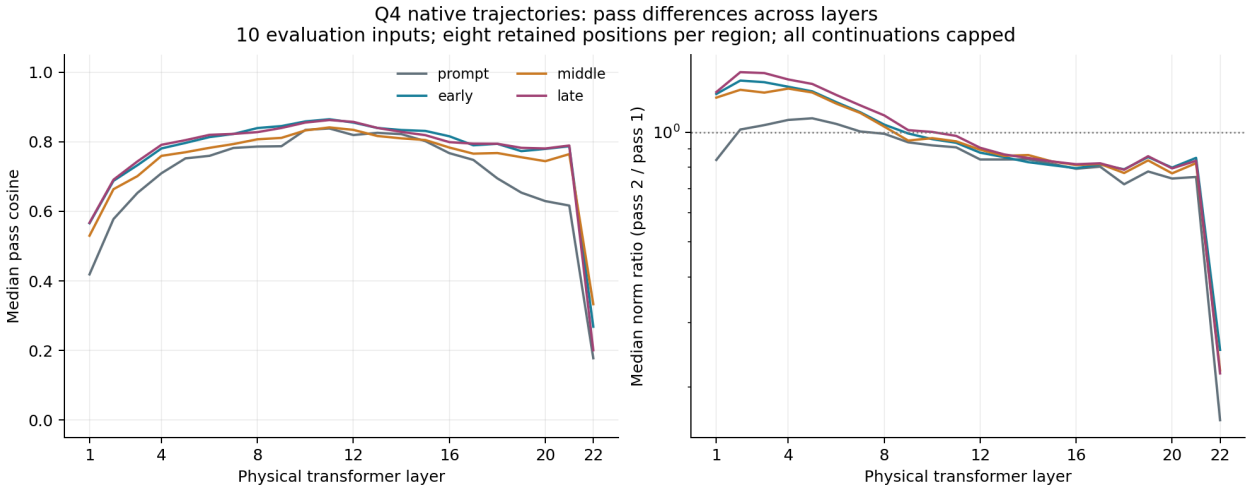
The 64-token GPU callback control produced exactly identical full-vocabulary logits with capture enabled and disabled. Conditional greedy checks during chunked trajectory replay matched 428/432 sampled next-token choices. The four mismatches had original-token logit gaps of approximately0.10-0.15. These are teacher-forced activations of the saved token sequences; they are not a claim that chunked replay reproduces every autoregressive choice. An initial process-level OpenMP conflict was resolved by separating a pure token-position helper from PyTorch imports, without a duplicate-runtime override.

The replay also produced 36 coordinate-transfer archives using the prior frozen BF16 VPD bases at physical 12/22. These store actual input projections, pass differences and basis-output differences, while omitting the residual outside the sparse basis. They do not constitute a newly fitted reasoning VPD or causal answer effects. Three shared SAEs have completed fitting on 256 retained training positions per pass from eight calibration inputs. Independent verification reproduced all 36 VPD coordinate archives, all 54 SAE feature-difference archives, and six held-out reconstruction measurements after reloading the dictionaries. All 900 activation arrays passed source-hash, shape, finite-value and position checks. The arithmetic

verification does not validate semantic feature interpretations. See `q4_reasoning_pilot_v1/mechanistic_verification.json`.

A descriptive region analysis contains 1,656 hidden-state comparisons (18 inputs, 23 sites, four regions) and 144 frozen-VPD region comparisons (18 inputs, two sites, four regions). Each region contains eight retained token positions. Among the ten evaluation input records, median pass cosine at physical layer 12 ranges from 0.819 to 0.857 across regions. At physical layer 22 it ranges from 0.178 to 0.333, and median second/first activation norm ratios range from 0.162 to 0.253. The sharp layer-22 change is present in the prompt suffix as well as the generated continuation. These are medians over input records, including multiple inputs from the same ARC task; they are not independent task-level estimates.

The callback sites are the residual outputs before loop-boundary normalization. At the separately recorded normalization site, median norm ratios instead range from 5.20 to 6.14. This site dependence cautions against equating a smaller residual norm with information compression. The pinned graph places both physical-layer output callbacks before their corresponding normalization operations. Pinned Nanbeige graph implementation.



Pass cosine and norm ratio across all physical layers and four trajectory regions

The generated-trajectory SAE at physical layer 12 yields held-out explained variance 0.2961 and 0.2511 for the two passes, versus training values 0.9832 and 0.9840. At physical layer 22, held-out values are 0.8662 and 0.0117, versus training values 0.9840 and 0.9381. These generalization gaps limit feature interpretation. Equal numbers of training rows do not equalize relative reconstruction loss: first-pass training variance at layer 22 is 5.62 times second-pass variance. This is a possible objective-scaling confound, not a demonstrated cause of the held-out gap; no objective has been changed in these fits. At loop-boundary normalization, held-out explained variance is 0.2496 and 0.4648, versus training values 0.8438 and 0.9939. All six held-out values were reproduced after independently reloading the saved dictionaries.

Shared SAE site	Training EV, pass 1	Training EV, pass 2	Held-out EV, pass 1	Held-out EV, pass 2
Physical layer 12	0.9832	0.9840	0.2961	0.2511
Physical layer 22	0.9840	0.9381	0.8662	0.0117
Loop-boundary normalization	0.8438	0.9939	0.2496	0.4648

The dictionaries remain shared across passes, with no evaluation fitting or pass-specific rescaling. The layer-22 second-pass reconstruction in particular is inadequate for strong feature-level claims.

4.8 Fresh external-controller controls

The empty archived logic bank after public-input exclusion was replaced, for a new controller-only experiment, by 128 separately constructed synthetic support puzzles. Another 32 synthetic puzzles were reserved for development and never entered retrieval. Generation uses seed 20260906 and eight fixed rectangular shapes from 4x4 to 7x6. Every constructed solution passes the official verifier and the literal per-tent adjacency check. All exact, rotated, and reflected aliases of the 109 public source puzzles are excluded, as are aliases within the synthetic set. The public set contains 108 distinct symmetry orbits. This construction favors puzzles satisfying both adjacency contracts and is not the publisher's generator or a reconstruction of historical training.

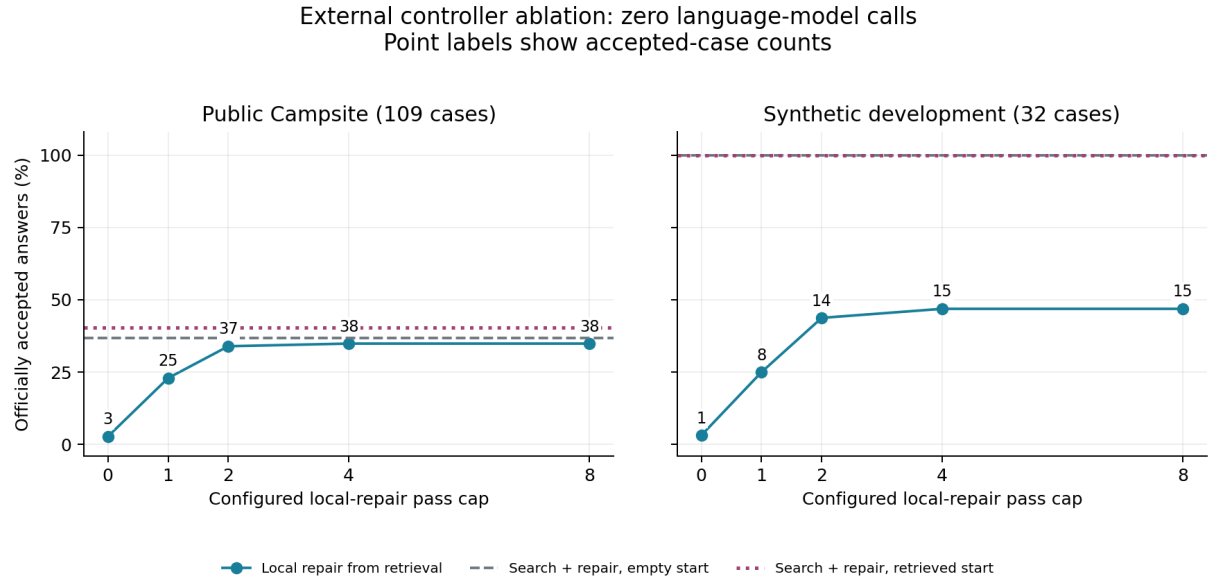
We ran the unmodified archived retriever and constraint corrector on the 32 synthetic development cases and all 109 public Campsite cases. These controls make zero language-model calls and do not constitute a full 27B+skill/TRM run. Retrieval supplies the nearest support solution as an initial candidate. Local repair first preserves the query's trees and dimensions, then performs up to the configured number of improving cell-sweep passes. Thus a zero-pass normalized initial grid differs from returning the raw retrieved answer. Search-enabled repair first attempts the archived exact routine and can fall back to local repair with an eight-pass cap and its original 200,000-node search bound.

Controller-only arm	Public official acceptance	Synthetic development acceptance
Raw retrieved solution	0/109	0/32
Normalize retrieved grid, zero repair passes	3/109	1/32
Local repair, cap 1	25/109	8/32
Local repair, cap 2	37/109	14/32
Local repair, cap 4	38/109	15/32
Local repair, cap 8	38/109	15/32
Search-enabled repair, empty start	40/109	32/32
Search-enabled repair, retrieved start	44/109	32/32

A separate validator using bit-mask dynamic programming for tent/tree matching reproduced all 1,128 saved official, literal and canonical outcome records. Literal adjacency and official matching are evaluated separately: two tents can each touch exactly one shared tree while failing a perfect matching. Four focused tests cover rectangular symmetry transformations, reproducible valid construction, and matching counterexamples. Original controller code and outputs remain unchanged.

Python call tracing then reproduced all 987 repair outputs exactly while observing actual execution. Search returned a solution on nine public cases and all 32 synthetic development cases, regardless of initial candidate. The remaining 100 public cases used local fallback. With search disabled and an eight-pass cap, public cases used one to five repair sweeps: counts 5, 57, 39, 7, and 1 respectively. The four- and eight-pass arms have identical final outputs on all 109 public cases and all 32 development cases. This is an observed early-stopping property of the external symbolic controller, not evidence about transformer routing.

Aggregate improvements can conceal individual regressions. Raising the local cap from two to four passes repairs two previously incorrect public cases but breaks one previously correct case. Changing the search-enabled initial candidate from empty to retrieved produces 13 gains and nine losses, for a net four-case increase. These paired controls identify effects within this particular controller and synthetic support bank. They do not isolate the historical language model's contribution or establish a neural recurrence mechanism. The perfect development acceptance of search-enabled repair also does not transfer to the public distribution.



Verified external-controller performance across configured repair caps

Artifacts are in `hermes_synthetic_campsite_v1` and `hermes_controller_control_v1`, including source copies, every answer, independent verification, paired transitions, and output-preserving execution traces. No neural controller was trained, and no new 27B inference was run in these controls.

4.9 BF16 generated-prefix VPD and pass-specific effects

The revised calibration completed on the exact 18 native Q4 trajectories, with eight training and ten held-out inputs. Its 64-component bases reconstruct 2.1919% and 2.1514% of squared weight energy at physical layers 12 and 22. Most weight energy remains in the residual. The increase from the earlier eight-component calibration is descriptive: both component count and calibration data changed, so this is not a controlled capacity scaling experiment.

All 20 held-out component-0 intervention records were retrieved and independently checked: 2,040 saved difference, interaction and causal-order calculations; 840 SAE re-encodings; and six held-out reconstruction measurements. A separate provenance check matches checkpoint assets, executed source, exact trajectories, 24 training prefixes and 50 joint updates to the frozen input contract. These checks verify saved arithmetic and lineage; they do not rerun the model interventions or validate feature semantics.

Site	SAE held-out EV, pass 1	SAE held-out EV, pass 2
Physical layer 12	0.0035	-0.0087
Physical layer 22	0.6667	-2.6589
Loop normalization	-0.0774	0.1381

These dictionaries were fitted to 64 training rows per pass, retaining the final eight positions of each training trajectory. They use the same 6,144-feature, top-32 architecture and 1,000-update budget. Several held-out reconstructions are worse than the held-out mean, particularly layer 22's second pass. The saved sparse-feature differences therefore do not support semantic mechanism claims. This fit also differs from Section 4.7 in precision and retained position coverage, preventing attribution of the quality change to one factor.

At coefficient +0.05, pooled final-normalization, second-pass hidden-state effect norms are:

Perturbed matrix	First only	Second only	Both passes	Interaction
Physical layer 12	75.7012	15.7032	75.6625	22.2407
Physical layer 22	31.7039	2.7813	31.7484	2.9706

Each row pools the same 80 held-out token positions. Negative doses and central differences are also retained in the raw archives. These are finite-dose hidden-state effects; intervention position changes downstream depth, and BF16 rounding can contribute to apparent interaction. They are not answer-success effects or evidence of adaptive routing. Baseline pooled pass cosine is 0.8530 at layer 12 and 0.2771 at layer 22; the second/first norm ratio is 0.9069 and 0.2096 respectively, then 5.4779 after normalization. The dependence on capture site again prevents a norm-only compression interpretation.

The new basis was also transferred to the previously verified Q4 down-projection inputs at both passes, yielding 36 coordinate and basis-output difference archives over 18 trajectories and both physical matrices. Exact full-token identity, split alignment and source hashes were checked. Float64 recomputation gives maximum relative coordinate error 3.86e-7. This is a projection of saved Q4 activations into the frozen BF16 basis, not direct BF16 input capture or a Q4 parameter fit; the residual contribution is omitted from these diagnostic outputs.

Run: `prime-20260906T204058Z-561`; manifest SHA256 `7c5205eeb3c1e7c025ac1aababc6e0526bbe82048f633a52e73bfefa07c8a546`; factors SHA256 `a1acd3ffa1fe0c4f4d2566b5e95670112bb152b31a149c05777b46ee6b4a502c`. The archive passed retrieval checks and the paid pod terminated successfully. `q4_reasoning_vpd64_transfer_v1` contains the coordinate transfer. The associated PNG/PDF/SVG figure records its source hashes and was visually reviewed.

4.10 Exploratory reconstruction diagnostics

Following the poor held-out SAE reconstruction, we fitted dense linear projection controls using only the original BF16 training activations. Each site supplies 64 rows per pass, giving 128 pooled rows and a centered training rank of 127. Shared-mean PCA controls retain ranks 0, 8, 32, 64 and 127; all ranks are reported in the diagnostic archive, without choosing a rank on held-out performance. Separate per-pass training means are also retained as controls. The table shows the saved SAE and two of the recorded projection ranks.

Site and pass	Saved SAE EV	Training PCA rank 8 EV	Full training-span EV
Layer 12, pass 1	0.0035	0.0008	0.1636
Layer 12, pass 2	-0.0087	0.0003	0.1705
Layer 22, pass 1	0.6667	0.9360	0.9470
Layer 22, pass 2	-2.6589	0.2044	0.3593
Loop norm, pass 1	-0.0774	-0.0976	0.1423
Loop norm, pass 2	0.1381	0.2692	0.4092

At layer 22's second pass, the training span supports substantially better reconstruction than the saved SAE achieves. This shows that the observed SAE failure is not explained solely by the portion of held-out activations outside that span. Dense PCA does not match the SAE's sparsity, feature count or semantic aims, and it is not a replacement mechanistic interpretation. Full-span projection still leaves substantial unexplained variance at most sites. An independent QR construction from training rows reproduces all 12 train/evaluation pass-level full-span reconstruction errors; all 12 saved SAE reconstruction errors were also recomputed. The main diagnostic additionally checks 72 projection/residual energy identities and stores all bases and reconstruction arrays.

First-pass training variance at layer 22 is 7.3216 times second-pass variance. Equal sampled row counts therefore do not equalize relative reconstruction error in the raw-MSE objective. A completed exploratory CPU control compares raw MSE with inverse-training-variance weighted MSE across three fixed seeds (42, 43, 44). Each fit retains 6,144 features, top-32 sparsity, 1,000 updates, batch size 128, the same pooled training-mean initialization, and AdamW at learning rate 0.001. Each seed uses identical initial parameters and sampled batches across the two arms; the two pass weights are 0.2403384 and 1.7596616, normalized to mean one. All six conditions are reported on the same training and held-out inputs:

Seed / objective	Training EV, pass 1	Training EV, pass 2	Held-out EV, pass 1	Held-out EV, pass 2
42 / raw MSE	0.9827	0.9505	0.6959	-2.6407
42 / weighted MSE	0.9729	0.9861	0.7068	-2.9388
43 / raw MSE	0.9957	0.9574	0.7192	-2.5289
43 / weighted MSE	0.9886	0.9851	0.7586	-3.0489
44 / raw MSE	0.9950	0.9639	0.7653	-2.7166
44 / weighted MSE	0.9860	0.9851	0.7730	-3.0866

Weighting improves training second-pass reconstruction in all three pairs but worsens held-out second-pass reconstruction in every pair. The paired held-out EV changes are -0.2981, -0.5199 and -0.3699 for seeds 42, 43 and 44, respectively. All six second-pass held-out EVs remain negative. Thus this particular variance-weighted objective does not repair the observed generalization failure; the control does not exclude other objective choices, optimization budgets or sources of mismatch. The three seeds do not supply a new independent evaluation cohort, and no feature semantics or answer-level benefit is established.

A separate NumPy float64 implementation recomputes encoder activations, top-k selection, decoder reconstructions and saved pass differences for all six dictionaries. It verifies all 108 feature archives and 24 pooled train/evaluation reconstruction measurements, together with source hashes, split identities, regenerated batch indices, paired initialization receipts, and all 1,000 recorded loss reductions per condition. These checks validate saved arithmetic and provenance without rerunning training. The complete receipt is `sae_loss_balance_verification/complete.json`.

We also align BF16 and Q4 captures at exactly the same final eight token positions on all 18 trajectories. On the ten held-out inputs, relative activation difference norms (Q4 minus BF16, divided by the BF16 norm) are 0.1363/0.1676 at layer 12, 0.0851/0.2623 at layer 22, and 0.1645/0.2287 at loop normalization for passes 1/2. These are pooled activation differences, not answer divergence rates. Quantization, numerical precision and implementation/runtime differences are not isolated by this comparison. The 108 per-input/site/pass records and split-level summaries are retained.

Artifacts: `sae_generalization_diagnostic_v2`, including `qr_verification.json`; completed paired control: `sae_loss_balance_control_v1`. These analyses were motivated by observed failures and are explicitly post-hoc. They do not establish new held-out generalization claims after model selection, feature semantics, or an answer-level recurrence benefit.

4.11 Signed response and output-proxy resolution

For each saved hidden-state pair at coefficients ±0.05, define the odd response as `(h_plus - h_minus) / 2` and the even response as `(h_plus + h_minus - 2*h_baseline) / 2`. These exactly reconstruct both signed changes in 720 arithmetic checks across 360 example/matrix/site/pass/scope records. A linear sign-reversing response would have zero even component. At final normalization after the second pass, pooled even/odd norm ratios are:

Perturbed matrix	First-pass-only intervention	Second-pass-only intervention	Both-pass intervention
Physical layer 12	1.5083	1.6505	1.4906
Physical layer 22	2.5569	0.7619	2.5268

The even component is comparable to or larger than the odd component in most of these conditions. Thus the saved coefficient is not demonstrated to lie in a linear-response regime. Nonlinearity and numerical execution effects are not isolated by this single-magnitude experiment. All sampled hidden-state values are exactly representable in BF16; at layer 22's second-only intervention, 47.58% of changed final-state coordinates move by no more than one larger adjacent BF16 step. This is a representational scale, not a measured noise floor. Smaller-dose, repeated-zero-dose and higher-precision model controls remain required before assigning an observed response to a stable component mechanism.

Every effect archive also contains full next-token vocabulary logits for baseline and all six signed interventions. Repeated baseline logits exactly match the separate baseline capture, and stored reference-token IDs/logit changes agree with those arrays. The original float32 KL proxies contain 34 negative values among 140 entries (including 20 baselines), with a minimum of -8.94e-8. Those negative values are numerical error. Float64 recomputation uses an equivalent generalized-KL expression to avoid cancellation near equality and agrees with direct normalized log-probability KL within 1e-12 absolute plus 1e-8 relative tolerance; original proxies remain unchanged.

Across the 120 non-baseline interventions, corrected KL ranges from 2.86e-16 to 0.02634, with median 1.54e-7; 68 values are below 1e-6 and 18 exceed 1e-3. None changes the highest-probability next-token ID. These distributions condition on one saved, budget-censored trajectory endpoint per input. They neither prove a lack of later behavioral effects nor demonstrate a final-answer improvement. Artifacts and full per-record values are in `signed_effect_resolution_v2`.

5. Prospective tests and limits

The next confirmatory stage should evaluate the full public ARC evaluation tasks and a fixed, family-stratified INTELLECT holdout; preserve all prompt-length exclusions and censoring; and compare paired outcomes under both equal-token and equal-compute budgets. A separate comparison should run the same questions through a pinned Hermes/TRM system with an auditable retrieval corpus and full tool-call costs. A solver-assisted arm must be labeled distinctly from a language-model-only arm.

For a fresh 27B comparator, we have pinned official Qwen3.5-27B metadata and a community Q4_K_M file by revision and SHA256. The weights have not been downloaded or evaluated. Its 64-layer hybrid architecture combines Gated DeltaNet and gated attention, so Nanbeige's layer-token execution proxy is not a portable compute measure across these models. Equivalence to the historical quantized checkpoint is unverified. See `comparator_sources/qwen35_27b/model_lock.json`. Official model description, pinned quantized checkpoint.

The fresh comparator now has verified model-facing inputs for both the 18-record pilot and all 1,223 fixed-cohort records. The full Qwen input set contains 2,503,714 prompt tokens, with a maximum of 17,609. Every original question field is retained; the publisher's native template trims only boundary whitespace on 31 model-facing prompts, and every rendered prompt round-trips through the pinned tokenizer. The inference bundle excludes grading sidecars. The prepared runner requires actual GGUF tokenizer agreement for every input, full GPU offload, exact weight/runtime hashes, and a saved-prefix check on resume. These controls have not yet been exercised on the 27B weights; no fresh 27B inference or TRM comparison score is reported. Independent input-identity checks are in `comparator_sources/qwen35_27b/input_identity_verification.json`.

Mechanism claims require stable component coverage, multiple SAE and decomposition seeds, a dose sweep, higher-precision controls, and causal interventions tied to final-answer success. Activation or feature norms alone do not prove reasoning compression. Larger or smaller trained architectures are required to estimate non-linear scale effects; changing inference loop count on one checkpoint is insufficient. Neither fixed Nanbeige recurrence nor an external Hermes router is evidence of an internal Astra router.

For prospective retrieval controls, the archived banks have now been filtered against the union of the fixed pilot and benchmark public inputs, preserving the evaluation cohorts. This removes all 100 logic-family support rows, leaving zero retriever or critic examples. Six math support rows remain, with no exact public-input match to these fresh cohorts; their overlap with the separate historical math evaluation remains documented. `hermes_support_fresh_v1/manifest.json` records source and output hashes and exclusion IDs. A separate synthetic logic bank and controller-only experiment now address this preparation gap as described in Section 4.8; they do not reproduce historical training or complete the model-assisted comparison.

The prepared answer-intervention stage starts fresh generations from each original held-out public prompt, rather than continuing the saved Q4 reasoning. It compares a zero-intervention baseline with positive and negative component-0 updates at each of physical layers 12 and 22, applied to the first pass, second pass or

both. At the initial coefficient magnitude 0.05 this gives 13 conditions and 130 generations on ten held-out inputs. Interventions cover all prefill and decode calls, so this stage does not isolate changes made during generated reasoning alone. Invocation counters reset at every model call, including cached decode steps; each condition uses a fresh KV cache. Seven controls on the publisher's small random recurrent architecture cover agreement with a physical shared-weight edit, full-prefix versus cached decoding, exact greedy-token agreement, causal ordering and restoration after an exception. These are implementation controls, not pretrained answer results. The pretrained stage requires completed VPD artifact and source-lineage verification, applies the unchanged cache-parity gate to all conditions, and grades re-decoded final answers locally with complete ARC-task aggregation. Dose and seed stability remain subsequent tests.

Reproducibility

Source lock: `sources.lock.json`. Dataset: `dataset_v1/{questions,graders}.jsonl` and `manifest.json`. Subset: `pilot_v1`. Solver audit: `hermes_audit/{summary.json,rows.jsonl}` with committed source copies and hashes. Experiment implementation: `../../experiments/nanbeige_reasoning/`; component and SAE implementation: `../../experiments/nanbeige_looped/`. The dependency files and tests are part of the experiment. This draft has not been submitted or published.